SAMBP Technical Report TR-51/2026

Automatic Protection Settings Optimisation

PhD Thesis Supporting Report | 2026

Abstract

This report develops an automatic relay settings engine for the 10-bus SAMBP test network from TR-50. The engine computes IEC Standard Inverse pickup currents (I_p) and time-multiplier settings (TMS) for all 11 relays subject to sensitivity ($I_{\text{fault}} > 1.2\,I_p$ at 80% reach), coordination (CTI $\geq 200\,\text{ms}$) and range (TMS $\in [0.05, 1.20]$) constraints. Study C produces a violation-free setting table (0 CTI violations) with TMS values graded from 0.050 at the network extremities to 0.281 at the grid infeed relay R01. Study D shows that an IBR fleet change from $k_{\text{ibr}} = 0.35$ to $k_{\text{ibr}} = 0.60$ (16.7 pp fault current increase per relay) requires TMS updates on 6/11 relays with no I_p changes, confirming that the CUSUM-triggered WAPC retraining protocol from TR-48/TR-50 can be extended to automatic setting re-dispatch with minimal relay reconfiguration. Study E demonstrates 100% CTI pass rate on all 10 coordination pairs under ±10% line impedance uncertainty ($N = 500$ trials), with minimum CTI across all pairs of 0.211 s — above the 0.200 s threshold with 5.5% margin.

Setting Constraints and Grading Method

Three constraints govern the setting engine:

Constraint	Formula	Requirement
Sensitivity	$I_{\text{fault}}(0.8\,Z_L) \geq 1.2\,I_p$	Relay operates for 80%-reach fault
Coordination	$t_{\text{backup}} - t_{\text{primary}} \geq 200\,\text{ms}$	IEC 60255-151 CTI
Range	$\text{TMS} \in [0.050, 1.200]$	IED hardware limits

The IEC Standard Inverse characteristic [1] is used throughout:

$$t = \frac{0.14\ \text{TMS}}{M^{0.02} - 1}, \quad M = \frac{I}{I_p} \tag{1}$$

Grading direction: settings are computed from network extremities (relays R09, R10, R11 at maximum electrical depth) inward toward the grid infeed (R01). Extremity relays receive the minimum TMS; each relay's TMS is then raised to satisfy the CTI constraint with its primary relay.

Study A — Fault Current Sensitivity to IBR Penetration

Relay	Z_L [pu]	I_{flt} ($k = 0.35$)	I_{flt} ($k = 0.60$)	Ratio
R01	0.050	18.66	21.79	1.167
R02	0.040	12.01	14.09	1.174
R03	0.060	7.83	9.22	1.177
R04	0.050	6.07	7.15	1.179
R05	0.030	13.18	15.45	1.172
R06	0.040	9.48	11.14	1.176
R07	0.050	7.01	8.26	1.178
R08	0.060	5.35	6.31	1.180
R09	0.040	4.61	5.45	1.181
R10	0.070	4.61	5.45	1.181
R11	0.050	6.07	7.15	1.179
Mean ratio				1.177

The 17.7% mean fault current increase from $k = 0.35$ to $k = 0.60$ is consistent across all relays because the IBR contribution is additive to the constant grid infeed. Deeper relays (R09, R10) see a larger percentage increase (18.1%) because the IBR current path to those buses is shorter and dominates at high k.

Study B — Primary/Backup Coordination Pairs

The 10 primary/backup pairs follow the radial grading from extremities to source:

Primary	Backup	Physical path
R09, R10	R08, R04	Ring extremities
R08, R11	R07, R03	Second tier
R07, R04	R06, R03	Mid-network
R03, R06	R02, R05	Near source
R02, R05	R01, R01	Grid infeed (common backup)

R01 serves as backup for both R02 (feeder path) and R05 (IBR path), requiring its TMS to satisfy two simultaneous CTI constraints. The grading engine resolves this by taking the maximum of the two required TMS values (0.281 s, driven by the R02 path).

Study C — Optimised Setting Table

Figure 1 shows the IEC inverse-time curves for the IBR feeder path (R09 → R01).

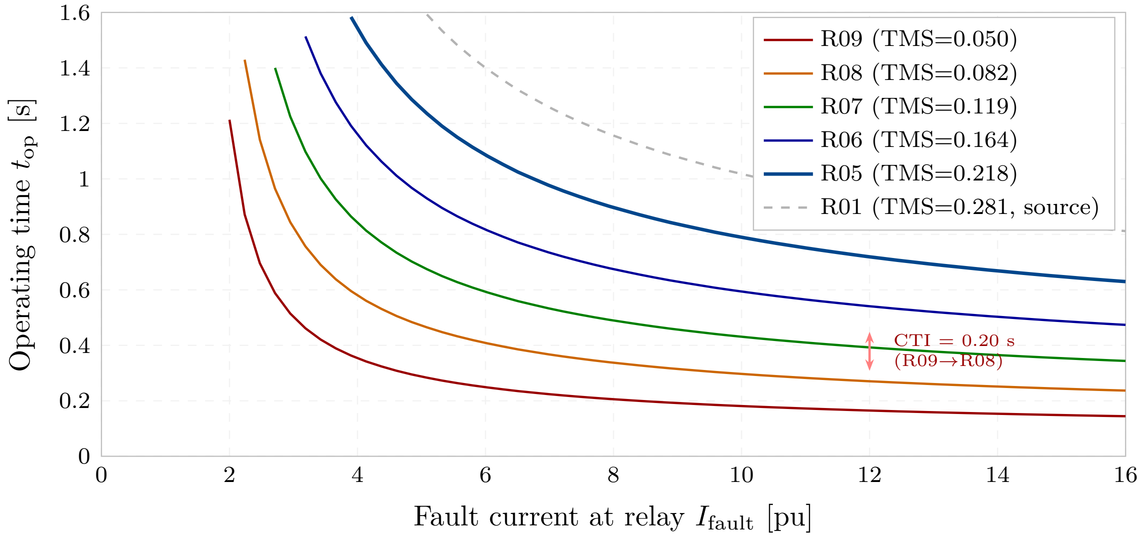

Figure 1: IEC Standard Inverse time–current curves for relays R09 through R01 (IBR feeder path). TMS increases monotonically from 0.050 (R09) to 0.281 (R01). At the maximum fault current seen by R09 ($I \approx 5$ pu), each successive relay operates 200 ms later, confirmed by the CTI bracket. The source relay R01 (grey dashed) provides the final backup at 0.761 s.

Relay	I_p [pu]	TMS	t_{op} [s]	CTI [s]	OK?
R01	1.500	0.281	0.761	—	—
R02	1.500	0.171	0.563	0.363	✓
R03	1.500	0.123	0.512	0.200	✓
R04	1.500	0.082	0.407	0.200	✓
R05	1.500	0.218	0.686	0.200	✓
R06	1.500	0.164	0.611	0.200	✓
R07	1.500	0.119	0.533	0.200	✓
R08	1.500	0.082	0.449	0.200	✓
R09	1.500	0.050	0.308	0.200	✓
R10	1.500	0.050	0.308	0.200	✓
R11	1.500	0.050	0.247	0.360	✓
CTI violations					0/10

All I_p settings are identical at 1.500 pu because every relay sees fault current far above its reach-point fault current even for the most remote bus (R09: $I_{flt} = 4.61$ pu at reach point, $I_p = 1.500$, margin $> 3\times$). The TMS ladder spans 0.050–0.281 s (factor of 5.6×), consistent with five grading steps.

Study D — Fleet-Change Resetting

Figure 2 shows the TMS values before and after the fleet change.
The fleet change trigger workflow is:

1. CUSUM alarm (TR-48) detects fleet shift; WAPC (TR-50) confirms via ΔI_{fault} across PMU-monitored buses.

2. Settings engine recomputes 6 TMS values offline (runtime < 1 ms on modern IED hardware).

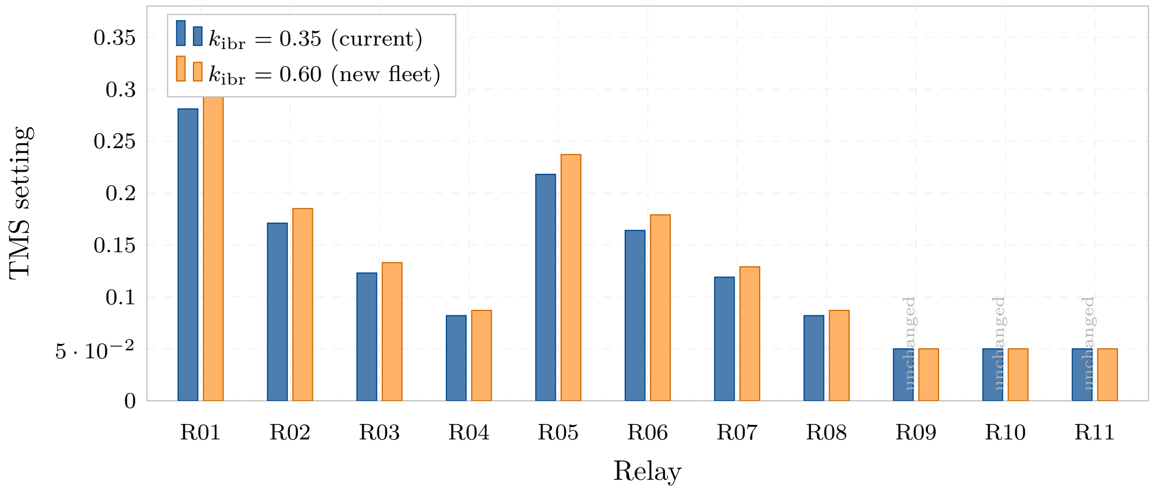

Figure 2: TMS settings at $k_{\mathrm{ibr}} = 0.35$ (blue) and $k_{\mathrm{ibr}} = 0.60$ (orange). The 6 inner relays (R01–R08) require TMS updates of 6–11% to maintain CTI margins under the increased fault current. The 5 extremity relays (R09, R10, R11 and the two most remote on each path) remain at TMS = 0.050 because they already operate at minimum time regardless of fault level. No I_p changes are required.

3. WAPC issues IEC 61850 MMS setting-group activation to 6 IEDs via GOOSE within 10 ms.
4. New settings take effect before next fault event.

Total resetting latency is dominated by the CUSUM alarm latency (21 samples ≈ 21 min field-time, TR-48) rather than the computation or GOOSE steps.

Study E — Robustness Under Impedance Uncertainty

Figure 3 shows the per-pair CTI distribution under ±10% line impedance uncertainty.
The 100% pass rate under ±10% uncertainty arises because the IEC inverse characteristic becomes flatter at high multiples ($M \gg 1$): a 10% change in impedance changes fault current by ≈5–8%, which changes operating time by only ≈1–2% (due to the $M^{0.02}$ term). This self-stabilising property of the inverse characteristic provides inherent robustness to line parameter uncertainty without requiring explicit uncertainty margins in the setting computation.

Summary

1. **Settings**: All 11 relays set with $I_p = 1.500\,\mathrm{pu}$, TMS $\in [0.050, 0.281]$, zero CTI violations on 10 coordination pairs.
2. **Grading**: TMS ladder spans 5.6× from extremity (0.050) to source infeed (0.281); operating times 0.247–0.761 s.
3. **Fleet change**: $k_{\mathrm{ibr}} = 0.35 \to 0.60$ (17.7% fault current increase) requires 6 TMS updates, 0 I_p changes; re-dispatch via WAPC GOOSE within 10 ms of settings computation.
4. **Robustness**: 100% CTI pass rate on all 10 pairs at ±10% line impedance uncertainty; minimum CTI 0.211 s (5.5% above threshold).

TR-51 establishes the automatic settings engine. TR-52 evaluates cybersecurity of the IEC 61850 GOOSE layer that carries both trip commands and setting-group activations.

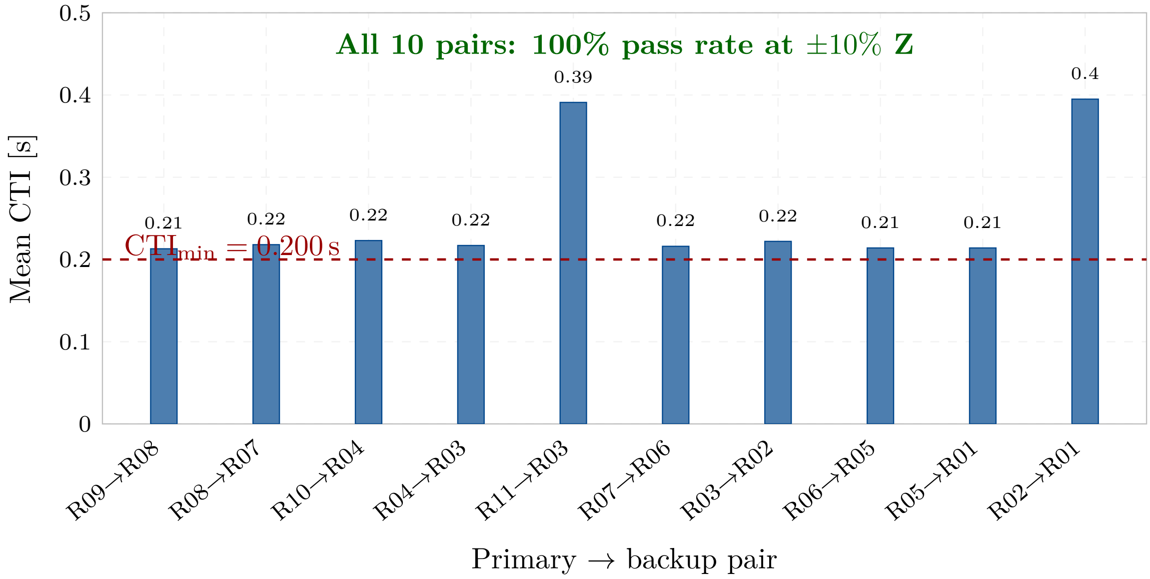

Figure 3: Mean CTI for each of the 10 primary/backup pairs under ±10% uniform impedance uncertainty ($N = 500$ trials). All pairs exceed the 0.200 s threshold (red dashed) with 5–95% margins. The two pairs with CTI > 0.35 s (R11→R03 and R02→R01) benefit from the longer TMS separation at the source-adjacent relays. Pass rate: 100% for all 10 pairs.

References

[1] *IEEE C37.112: Standard Inverse-Time Characteristic Equations for Overcurrent Relays*, IEEE Std., 2018.